

Section A — MCQs.

1. (b) 8.
2. (b) Contains all elements under consideration.
3. (a) $4/5$
4. (c) Many one relation.
5. (b) $4/2$
6. (c) 60
7. (b) $A \cap B = \emptyset$
8. (a) Reflexive.

Section B — Very short Answers.

- 9- A set is very well-defined collection of objects examples: $A = \{1, 2, 3\}$.
- 10- Domain of $f(x) = x^2$ is $\mathbb{R}$ and range is $[0, \infty)$.
- 11- $\sin 45^\circ \cos 45^\circ = (1/\sqrt{2})(1/\sqrt{2}) = 1/2$.
- 12- $5P2 = 5 \times 4 = 20$.
- 13- Number of elements in power set of A is $2^4 = 16$.
- 14- One-one function: $f(x) = x$; Many-one function: $f(x) = x^2$.

Section C — Short Answer.

15- $LHS = \sin \alpha / (1 + \cos \alpha) + (1 + \cos \alpha) / \sin \alpha$

Taking LCM and simplifying we get $RHS = 2 \operatorname{Cosec} \alpha$.

16- $A \cup B = \{1, 2, 3, 4, 5, 6\}$ and $A \cap B = \{3, 4\}$.

17- Total 3-digit numbers = $5P3 = 5 \times 4 \times 3 = 60$

(Reasoning briefly)

18- Relation is reflexive, symmetric and transitive.

Section D- Long Answers.

$$19 - (1 + \tan^2 \theta)(1 + \sin^2 \theta)$$

$$= \sec^2 \theta (1 + \sin^2 \theta)$$

Using identity $1 + \tan^2 \theta = \sec^2 \theta$.

Hence LHS = RHS.